Electromagnetic diffraction of light focused through a stratified medium

P. Török and P. Varga

We consider the focusing of light by a high-aperture lens into a stratified medium. The solution is based on our previously obtained results [J. Opt. Soc. Am. A **12,** 325 (1995)], where we represented the illumination incident upon a plane interface between media with mismatched refractive indices as a sum of plane waves. The present solution is obtained in terms of plane waves, and it satisfies Maxwell's equations. The diffraction integrals are obtained in a form that is readily computable. We present numerical examples for some practical cases. © 1997 Optical Society of America

1. Introduction

Light propagation through stratified media is of particular interest mainly because of holography investigations[1] and light-microscopy applications,[2] the former being less relevant to our present problem. In light and especially in confocal microscopy, stratified media are frequently used as specimens, e.g., integrated circuits, thin films, optoelectronic devices, etc. In this paper the primary objective is to determine the electromagnetic field inside a stratified structure and to examine how spherical aberration, introduced by variations along the optical axis in the refractive index, affects the localization of the probe.

The purpose of this paper is to extend our previous treatment of the diffraction problem[3] for the case when an electromagnetic wave is focused through an interface between media of mismatched refractive indices[4] to the case when light is focused by a high-aperture system into a stratified medium. The starting point of our treatment is Wolf's integral formulas,[5,6] with which we derive the electromagnetic field just before the interface between the first two media. The field is then traversed across the stratified medium by application of the Fresnel refraction law to the individual plane waves. The field so derived just after the interface is used as the boundary condition for a second set of integral formulas corresponding to a superposition of plane waves, which represent the field inside the Nth medium. In this way the diffraction problem is solved in a rigorous mathematical manner, and the solution satisfies Maxwell's equations.

The construction of the paper is as follows. First we derive the diffraction integrals in the Nth medium. We give a physical interpretation to our integrals, after which numerical results are presented on the distribution of the electric-energy density. These numerical results correspond to the case when the point of observation is at a certain depth below the last interface and the specimen is axially scanned with respect to the lens. We note that our procedure can directly be applied to the magnetic vector field, whose presentation is omitted for brevity.

2. Integral Representation

Consider an optical system of revolution with an optical axis z, as shown in Fig. 1. This system images a point source, which is situated in the object space at $z = -\infty$ and radiates a linearly polarized monochromatic and coherent electromagnetic wave. This wave is incident on a lens of aperture Σ, which produces an aberration-free convergent spherical wave in the image space. The origin O of the (x, y, z) coordinate system is positioned at the Gaussian focus. The electric field is determined at the arbitrary point P in the focal region. The aperture size and the distance of P from the aperture are taken to be large compared with the wavelength. In Fig. 1 and the following, $\hat{s}_N = (s_{Nx}, s_{Ny}, s_{Nz})$ is the unit vector along a typical ray in the Nth medium and $\mathbf{r}_p = (x, y,$

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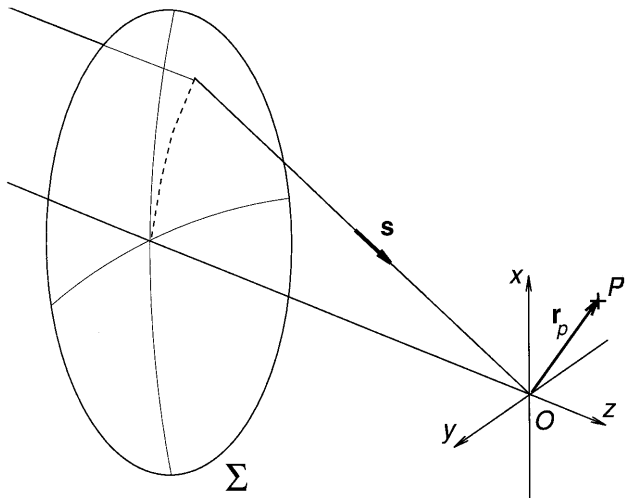


Fig. 1. Diagram showing light focused by a lens into a single medium.

z) is the position vector pointing from O to P. In the following we shall assume the usual time dependence of the electric field.

In a homogeneous image space, the time-independent electric field can be represented as a superposition of plane waves,[7] and we use the form developed by Wolf[5]:

$$\mathbf{E}(P) = -\frac{ik_1}{2\pi}\int\int_{\Omega}\frac{\mathbf{a}(s_{1x}, s_{1y})}{s_{1z}}\exp[ik_1\hat{s}_1 \cdot \mathbf{r}_p]\mathrm{d}s_{1x}\mathrm{d}s_{1y}, \tag{1}$$

where $\mathbf{a}$ is the electric strength vector of the unperturbed electric field in the exit aperture Σ, k_1 is the wave number, and Ω is the solid angle formed by all the geometrical optics rays.

The case corresponding to Fig. 1, but for an image space consisting of the stratified medium with wave numbers $k_1, k_2, \ldots, k_N$ is shown in Fig. 2. The origin O is again positioned at the Gaussian focus. We reformulate Eq. (1) as follows. In the first medium and at the first interface ($z = -h_1$), the incident

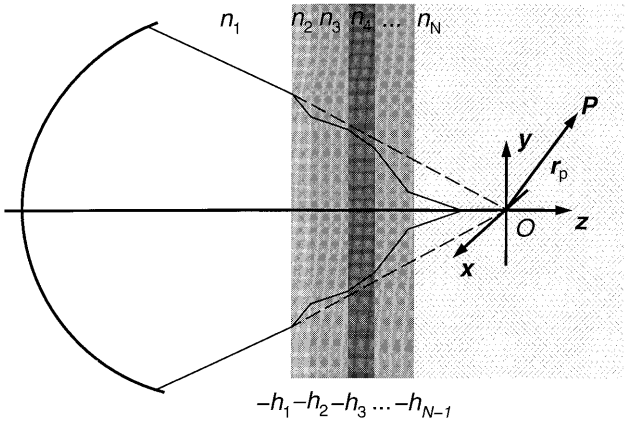


Fig. 2. Diagram showing light focused by a lens into two media separated by a planar interface.

electric field is given by

$$\mathbf{E}_1^{(1)}(x, y, -h_1) = -\frac{ik_1}{2\pi}\int\int_{\Omega_1}\frac{\mathbf{a}(s_{1x}, s_{1y})}{s_{1z}} \times \exp[ik_1(s_{1x}x + s_{1y}y - s_{1z}h_1)]\mathrm{d}s_{1x}\mathrm{d}s_{1y}. \tag{2}$$

To describe the field in the Nth medium we assume that each plane-wave component refracting at the interfaces obeys the Fresnel refraction law and the resulting field in the Nth medium is constructed as a superposition of refracted plane waves. If the amplitude of the plane waves incident on the interface is described by $\mathbf{W}(\hat{s}_1) = \mathbf{a}/s_{1z}$, then the amplitude of the transmitted plane waves in the Nth medium is a linear function of $\mathbf{W}(\hat{s}_1)$, i.e., $\mathbf{\Lambda W}$, where the tensor operator $\mathbf{\Lambda}$ is a function of the angle of incidence, the thickness of the individual layers, and $k_1, k_2, \ldots, k_N$. The transmitted field in the Nth medium, at the close vicinity of the last interface, is given by

$$\mathbf{E}_N(x, y, -h_{N-1}) = -\frac{ik_1}{2\pi}\int\int_{\Omega_1}\mathbf{\Lambda W}(\hat{s}_1) \times \exp[ik_1(s_{1x}x + s_{1y}y - s_{1z}h_{N-1})] \times \mathrm{d}s_{1x}\mathrm{d}s_{1y}. \tag{3}$$

We represent the field inside the Nth medium again as a superposition of plane waves. This representation is a solution of the time-independent wave equation and can be written as

$$\mathbf{E}_N(x, y, z) = -\frac{ik_N}{2\pi}\int\int_{\Omega_N}\mathbf{F} \times \exp[ik_N(s_{Nx}x + s_{Ny}y - s_{Nz}z)]\mathrm{d}s_{Nx}\mathrm{d}s_{Ny}. \tag{4}$$

We have to determine the function $\mathbf{F}$, and for this we shall make use of Eq. (3), which represents a boundary condition for Eq. (4). First, however, we need to establish the relation among $\hat{s}_1$, $\hat{s}_2$, and, in general, $\hat{s}_N$.

It is evident from the vectorial law of refraction,

$$k_2\hat{s}_2 - k_1\hat{s}_1 = (k_2\cos\theta_2 - k_1\cos\theta_1)\hat{u}, \tag{5}$$

where $\hat{u}$ represents the normal of the interface, $\theta_1 = (\hat{s}_1, \hat{u})$, and $\theta_2 = (\hat{s}_2, \hat{u})$, that

$$k_N s_{Nx} = \ldots = k_2 s_{2x} = k_1 s_{1x},$$

$$k_N s_{Ny} = \ldots = k_2 s_{2y} = k_1 s_{1y}. \tag{6}$$

On expanding Eq. (4) and expressing $\mathbf{F}$, as shown in

Ref. 3, we obtain the electric field in the Nth medium:

$$\mathbf{E}_N(x, y, z) = -\frac{ik_1}{2\pi}\int\int_{\Omega_1} \mathbf{\Lambda W}(\hat{s}_1) \times \exp[ik_0(n_N h_{N-1} s_{Nz} - n_1 h_1 s_{1z})] \times \exp(ik_N s_{Nz} z)\exp[ik_1(s_{1x}x + s_{1y}y)]\mathrm{d}s_{1x}\mathrm{d}s_{1y}. \tag{7}$$

From the above it also follows that in Eq. (7)

$$s_{lz} = \left[1 - \frac{{k_1}^2}{{k_l}^2}({s_{1x}}^2 + {s_{1y}}^2)\right]^{1/2}. \tag{8}$$

For numerical purposes, however, Eq. (8) cannot be applied directly. This is because the stratified medium can contain interfaces at which refraction is not ordinary (i.e., total reflection or generation of evanescent waves). We shall, therefore, use for numerical purposes a successive computation, i.e., $s_{1z} \rightarrow s_{2x} \rightarrow \ldots s_{Nz}$. It is important to emphasize that, since both the boundary condition represented by Eq. (3) and the integral representation Eq. (4) are exact solutions of Maxwell's equations, our formulas for the electric vector [Eq. (7)] in the Nth medium also satisfy Maxwell's equations. Therefore, we have successfully obtained a consistent extension of Wolf's solution in a stratified medium.

3. Electric Strength Vector

Determination of the electric strength vector for a single and double medium of propagation was described previously by Richards and Wolf[6] and Török *et al.*[3] Now we obtain these vectors for a plane polarized wave incident on the lens and a stratified medium. Individual layers of the stratified medium are taken to be isotropic and homogeneous and to have an optically smooth planar surface that is perpendicular to the optical axis. For our decomposition, the usual assumptions are made, namely, that the electric vector maintains its direction with respect to a meridional plane and the electric vector remains on the same side of a meridional plane on passing through the system.

For the optical system under consideration, the angle of incidence at the interface is denoted by θ_1 and the angle in the Nth medium by θ_N. The unit vector $\hat{s}_N$ and position vector $\mathbf{r}_p$ (Fig. 2) are given in spherical polar coordinates by

$$\hat{s}_N = \sin\theta_N \cos\phi\hat{\imath} + \sin\theta_N \sin\phi\hat{j} + \cos\theta_N\hat{k}, \tag{9}$$

$$\mathbf{r}_p = r_p(\sin\theta_p \cos\phi_p\hat{\imath} + \sin\theta_p \sin\phi_p\hat{j} + \cos\theta_p\hat{k}. \tag{10}$$

where $\hat{\imath}, \hat{j}$, and $\hat{k}$ are the unit base vectors of the (x, y, z) orthogonal system and the spherical polar coordinates r, θ, and ϕ are defined so that $r > 0$, $0 \le \theta < \pi$, and $0 \le \phi < 2\pi$. The coordinate system is chosen so that the y component of the incident electric vector is zero. We set for the incident electric vector $\mathbf{e}_0$ in front of the lens $\mathbf{e}_0 = (e_0, 0, 0)$. For a treatment of the refraction that occurs at the interface, it is convenient to decompose the electric vector into s- and p-polarized vector components, e_s and e_p, respectively, and to rotate the coordinate system so that the new coordinate system will contain components in the (p, s, ζ) system. This coordinate system is defined in such a way that $e_\zeta = 0$. The electric vector components $\mathbf{e}_{(p,s,\zeta)}$ after the lens are then in the following form:

$$\mathbf{e}_{(p,s,\zeta)} = A(\theta_1)\mathbf{P}^{(1)}\mathbf{LRe}_0, \tag{11}$$

where $A(\theta_1)$ is an amplitude function (defined below), the matrix $\mathbf{R}$ describes the coordinate transformation for rotation around the z axis:

$$\mathbf{R} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{12}$$

the matrix $\mathbf{L}$ describes the changes in the electric field as it traverses the lens:

$$\mathbf{L} = \begin{bmatrix} \cos\theta_1 & 0 & \sin\theta_1 \\ 0 & 1 & 0 \\ -\sin\theta_1 & 0 & \cos\theta_1 \end{bmatrix}, \tag{13}$$

and the matrix $\mathbf{P}^{(l)}$ describes the coordinate-system rotation that generates e_s and e_p components with $e_\zeta = 0$ in medium l:

$$\mathbf{P}^{(l)} = \begin{bmatrix} \cos\theta_l & 0 & -\sin\theta_l \\ 0 & 1 & 0 \\ \sin\theta_l & 0 & \cos\theta_l \end{bmatrix}. \tag{14}$$

The electric field in the Nth medium $\mathbf{e}_N$ is given by

$$\mathbf{e}_N = [\mathbf{R}]^{-1}[\mathbf{P}^{(N)}]^{-1}\mathbf{I}^{(N-1)}\mathbf{e}_{(p,s,\zeta)}, \tag{15}$$

where the matrix $\mathbf{I}$ describes the effect of the stratified medium:

$$\mathbf{I}^{(N-1)} = \begin{bmatrix} T_p^{(N-1)} & 0 & 0 \\ 0 & T_s^{(N-1)} & 0 \\ 0 & 0 & T_p^{(N-1)} \end{bmatrix}. \tag{16}$$

where $T_{s,p}^{(N-1)}$ is the transmission coefficient of the stratified medium describing the s- and p-polarized light traversing $N - 1$ media and is given in Appendix A:

$$T_m^{(N-1)} = \frac{t_m^{(N-1)} \prod_{j=1}^{N-2} t_m^{(j)} \exp(i\beta_{j+1})}{D_m^{(N-1)}}, \tag{17}$$

where $\beta_l = k_l(h_{l-1} - h_l)\cos\theta_l$, $m = p, s$, $D^{(N-1)}$ is given in Appendix A, and $t_p^{(l)}$, $t_s^{(l)}$ and $r_p^{(l)}$, $r_s^{(l)}$ are the

Fresnel coefficients for transmission and reflection, respectively[8]:

$$t_s^{(l)} = \frac{2n_l \cos\theta_l}{n_l \cos\theta_l + n_{n+1} \cos\theta_{l+1}},$$

$$t_p^{(l)} = \frac{2n_l \cos\theta_l}{n_{l+1} \cos\theta_l + n_l \cos\theta_{l+1}}, \tag{18}$$

$$r_s^{(l)} = \frac{n_l \cos\theta_l - n_{l+1} \cos\theta_{l+1}}{n_l \cos\theta_l + n_{l+1} \cos\theta_{l+1}},$$

$$r_p^{(l)} = \frac{n_{l+1} \cos\theta_l - n_l \cos\theta_{l+1}}{n_{l+1} \cos\theta_l + n_l \cos\theta_{l+1}}. \tag{19}$$

From Eqs. (11)–(13) and (15) we obtain the components of the electric vector in the Nth medium on setting $e_0 = 1$:

$$\mathbf{e}_N = A(\theta_1) \times \begin{bmatrix} T_p^{(N-1)} \cos\theta_N \cos^2\phi + T_s^{(N-1)} \sin^2\phi \\ T_p^{(N-1)} \cos\theta_N \sin\phi \cos\phi - T_s^{(N-1)} \sin\phi \cos\phi \\ -T_p^{(N-1)} \sin\theta_N \cos\phi \end{bmatrix}. \tag{20}$$

Function $A(\theta_1)$ can be regarded as an apodization function that depends on the lens used in imaging. Richards and Wolf[6] showed that, when the system obeys the Abbe sine condition, i.e., is aplanatic, then

$$A(\theta_1) = f l_0 \cos^{1/2}\theta_1, \tag{21}$$

where f is the focal length of the lens and l_0 is an amplitude factor. Having derived the electric vector $\mathbf{e}_N$, we can write the electric strength vector as

$$\mathbf{\Lambda W}(\hat{s}_1) = \mathbf{e}_N. \tag{22}$$

4. Electric-Field Vector

First we formulate the expressions needed to simplify Eq. (7). We transform the integral variables ds_{1x}, ds_{1y} to the spherical polar coordinate system and define

$$\kappa = k_1 \sin\theta_1 \sin\theta_p \cos(\phi - \phi_p), \tag{23}$$

$$\Psi_i = h_{N-1} n_N s_{Nz} - n_1 h_1 s_{1z}. \tag{24}$$

Ψ_i shall be referred to in the following as the initial aberration function. So far we have constructed the quantities required to determine the electric vector. With the help of the above results Eq. (7) can be rewritten to express the electric vector $\mathbf{E}_N$ in the Nth medium:

$$\mathbf{E}_N(\mathbf{r}_p) = -\frac{ik_1}{2\pi} \int\int_{\Omega_1} \mathbf{e}_N \exp(ir_p\kappa)\exp(ik_0\Psi_i) \times \exp(ik_N r_p \cos\theta_p \cos\theta_N)\sin\theta_1 d\theta_1 d\phi. \tag{25}$$

On substituting from Eqs. (22) and (20) into Eq. (25), changing the integration limits and assuming that the system obeys the sine condition [Eq. (21)], setting α to be the angular semiaperture of the lens, and carrying out the integration with respect to ϕ, we obtain the following expression for the electric-field components:

$$E_{Nx} = -iK(I_0^{(N)} + I_2^{(N)} \cos 2\phi_p),$$
$$E_{Ny} = -iKI_2^{(N)} \sin 2\phi_p,$$
$$E_{Nz} = -2KI_1^{(N)} \cos\phi_p, \tag{26}$$

where

$$K = \frac{k_1 f l_0}{2},$$

and the integrals $I_0^{(N)}$, $I_1^{(N)}$, and $I_2^{(N)}$ are given by

$$I_0^{(N)} = \int_0^\alpha \sqrt{\cos\theta_1} \sin\theta_1 \exp(ik_0\Psi_i) \times (T_s^{(N-1)} + T_p^{(N-1)} \cos\theta_N) \times J_0(k_1 \sin\theta_1 r_p \sin\theta_p)\exp(ik_N r_p \cos\theta_p \cos\theta_N)d\theta_1,$$

$$I_1^{(N)} = \int_0^\alpha \sqrt{\cos\theta_1} \sin\theta_1 \exp(ik_0\Psi_i) T_p^{(N-1)} \sin\theta_N \times J_1(k_1 \sin\theta_1 r_p \sin\theta_p)\exp(ik_N r_p \cos\theta_p \cos\theta_N)d\theta_1,$$

$$I_2^{(N)} = \int_0^\alpha \sqrt{\cos\theta_1} \sin\theta_1 \exp(ik_0\Psi_i) \times (T_s^{(N-1)} - T_p^{(N-1)} \cos\theta_N) \times J_2(k_1 \sin\theta_1 r_p \sin\theta_p)\exp(ik_N r_p \cos\theta_p \cos\theta_N)d\theta_1, \tag{27}$$

where J_n is the Bessel function of the first kind, order n. Equations (26) and (27) conclude our solution of the problem. It should be noted that, for $N = 1$, $k_1 = k_N = 1$ (therefore $\theta_N = \theta_1$, $h_j = 0$, and $d = 0$), Eqs. (27) reduce to a set that is identical to the corresponding equations of Richards and Wolf.[6] For $N = 2$, Eqs. (26) and (27) are simplified and yield the same result as our previous investigation for a single dielectric interface.[3] Therefore, our formulation of the vector diffraction theory for stratified media is consistent with previous results.

5. Physical Interpretation

It is not difficult to see that, as we showed for the problem of a single interface,[3] Eq. (7) represents the Fourier transform of a function consisting of a defocus term, the electric strength vector, and the aberration function. This is readily conceivable, as the present diffraction integral is essentially in the same form as our previously obtained diffraction integral,[3] with a small difference in the aberration function, this being more complex for the stratified medium.

It is, therefore, important to investigate the aberration function in more detail.

It is clear that in Eqs. (27) the aberration function consists of the initial aberration function [Eq. (24)] and the additional phase from $T_m^{(N-1)}$. As shown in Appendix A, $T_m^{(N-1)}$ possesses a denominator that is of rather complex structure. It is, however, possible, for certain cases, to consider this denominator as unity. For practical cases, primarily in the biological sciences, the specimen is constructed from layers with close refractive indices. For such layers the reflection coefficients are much smaller than unity. For example, consider a glass–water interface for $\alpha = 0.9$ rad. The reflection coefficients give 0.1 and 0.25 (for the s and p directions, respectively), whereas for a water–glass interface these values are 0.05 and 0.25, respectively. We shall therefore consider in the following part of this section the denominator of $T_m^{(N-1)}$ as unity.

By recalling Eq. (17),

$$T_m^{(N-1)} = \frac{t_m^{(N-1)} \prod_{j=1}^{N-2} t_m^{(j)} \exp(i\beta_{j+1})}{D_m^{(N-1)}},$$

we find that, from the term $\exp(i\beta_{j+1})$, the additional phase factor Ψ_a of the aberration function is given by

$$\begin{aligned}\Psi_a = {} & h_1 n_2 \cos\theta_2 + h_2(n_3 \cos\theta_3 - n_2 \cos\theta_2) \\ & + \ldots + h_{N-2}(n_{N-1} \cos\theta_{N-1} - n_{N-2} \cos\theta_{N-2}) \\ & - h_{N-1} n_{N-1} \cos\theta_{N-1}.\end{aligned}$$

The above equation results, by use of Ψ_i, in the overall aberration function:

$$\Psi = \sum_{j=1}^{N-1} h_j(n_{j+1} \cos\theta_{j+1} - n_j \cos\theta_j), \tag{28}$$

which can also be written as

$$\begin{aligned}\Psi = {} & -h_1 n_1 \cos\theta_1 + h_{N-1} n_N \cos\theta_N \\ & + \sum_{l=2}^{N-1} (h_{l-1} - h_l) n_l \cos\theta_l.\end{aligned} \tag{29}$$

Equation (29) means that the aberration function consists of individual terms that are responsible for the phase difference during propagation. It is shown that the phase of a ray in the jth medium is proportional to the thickness of the layer and to the longitudinal projection of the wave vector, as one might have anticipated. Expression (29) can be rewritten if we assume that there are no interfaces in the stratified structure that would cause nonordinary refraction:

$$\begin{aligned}\Psi = {} & -h_1 n_1 \sqrt{1 - \rho^2 \sin^2\alpha} \\ & + h_{N-1} n_N \left[1 - \left(\frac{n_1}{n_N}\right)^2 \rho^2 \sin^2\alpha\right]^{1/2} \\ & + \sum_{j=2}^{N-1} (h_{j-1} - h_j) n_j \left[1 - \left(\frac{n_1}{n_j}\right)^2 \rho^2 \sin^2\alpha\right]^{1/2},\end{aligned} \tag{30}$$

where we have defined the normalized radial coordinate $\rho = \sin\theta_1/\sin\alpha$. Individual terms in Eq. (30) can be expanded to give

$$\begin{aligned}\Psi = {} & -h_1 n_1 \sum_{l=0}^{\infty} \frac{1}{l!} a_l \rho^{2l} \sin^{2l}\alpha \\ & + h_m \sum_{l=0}^{\infty} \frac{1}{l!} a_l n_1 \left(\frac{n_1}{n_N}\right)^{2l-1} \rho^{2l} \sin^{2l}\alpha \\ & + \sum_{j=2}^{N-1} (h_{j-1} - h_j) \sum_{l=0}^{\infty} \frac{1}{l!} a_l n_1 \left(\frac{n_1}{n_j}\right)^{2l-1} \rho^{2l} \sin^{2l}\alpha,\end{aligned} \tag{31}$$

from which it is shown that, as $l \to \infty$, the higher-order aberration coefficients are independent of the wave numbers (refractive indices) of the individual layers in the stratified medium if $n_1 < n_j$, $j = 2, 3, 4, \ldots$.

6. Numerical Results

Undoubtedly the most important application of the above theory is in the biological sciences. Biological specimens are usually formed as two- or three-layer stacks of different refractive-index media. Typically dry or water- or oil-immersion objective lenses are used to focus light through the coverglass and to form the probe in the usually watery specimen. When oil-immersion lenses are used, there are practically only two media present (glass and water), and thus the previous theory[3] and our results[9,10] can readily be applied. When, however, water-immersion or dry objective lenses are used, the above theory should be applied to compute the electromagnetic field inside the embedding medium. For this reason we shall in the following confine our exact and general solutions to the case of a three-layer stratified medium.

For this case, by using Eq. (17), we can write:

$$T_{s,p}^{(N-1)} = \tau_{s,p} = \frac{t_{12}^{s,p} t_{23}^{s,p} \exp(i\beta)}{1 + r_{12}^{s,p} r_{23}^{s,p} \exp(2i\beta)}, \tag{32}$$

where $\beta = 2k_2|h_1 - h_2|\cos\theta_2$. The time-averaged electric-energy density w_e is given from Eqs. (26) by

$$\langle w_e(r_p, z, \theta_p)\rangle = \frac{1}{16\pi} (\mathbf{E}_N \cdot \mathbf{E}_N^*). \tag{33}$$

For the numerical evaluation of Eq. (33), C++ programs were written and Numerical Recipes[11] library routines were used. The programs were run on an

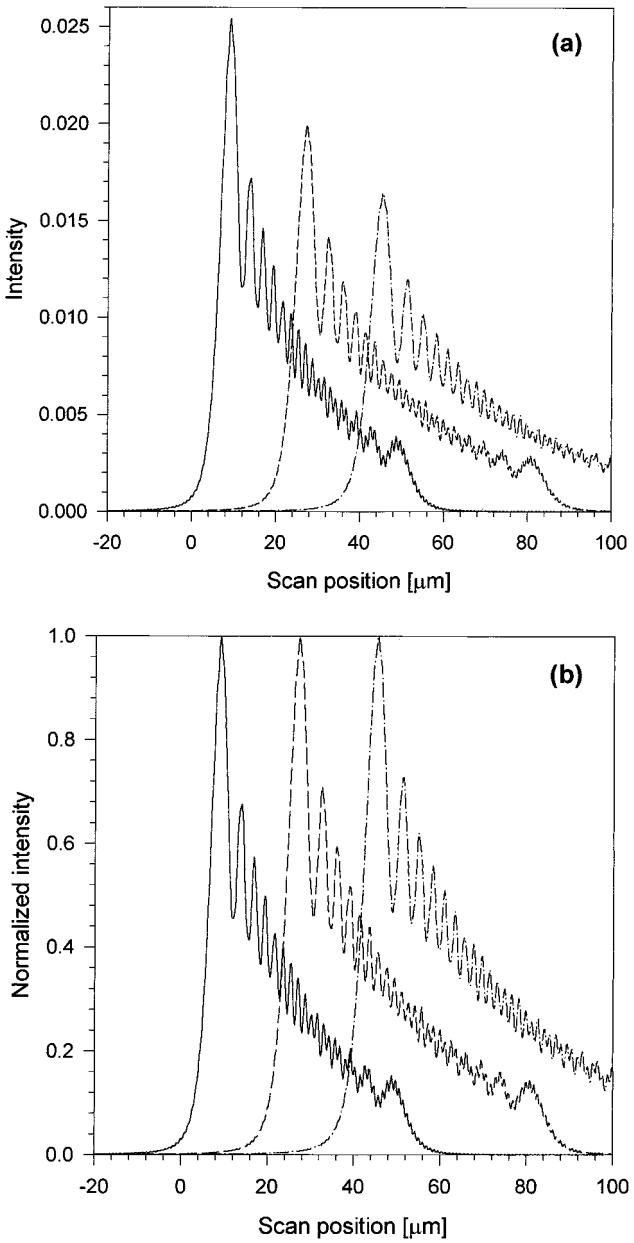


Fig. 3. Time-averaged electric-energy density distributions as functions of the scan position for a lens numerical aperture of 0.9 and for focusing from air ($n_1 = 1$) through a cover glass ($n_2 = 1.54$) into an aqueous medium ($n_3 = 1.33$). The thicknesses of the cover glasses are 120 (solid curve), 170 (dashed curve), and 220 μm (dashed-dotted curve) and the depth in the aqueous medium is 50 μm. The lens is not corrected for the cover-glass thickness.

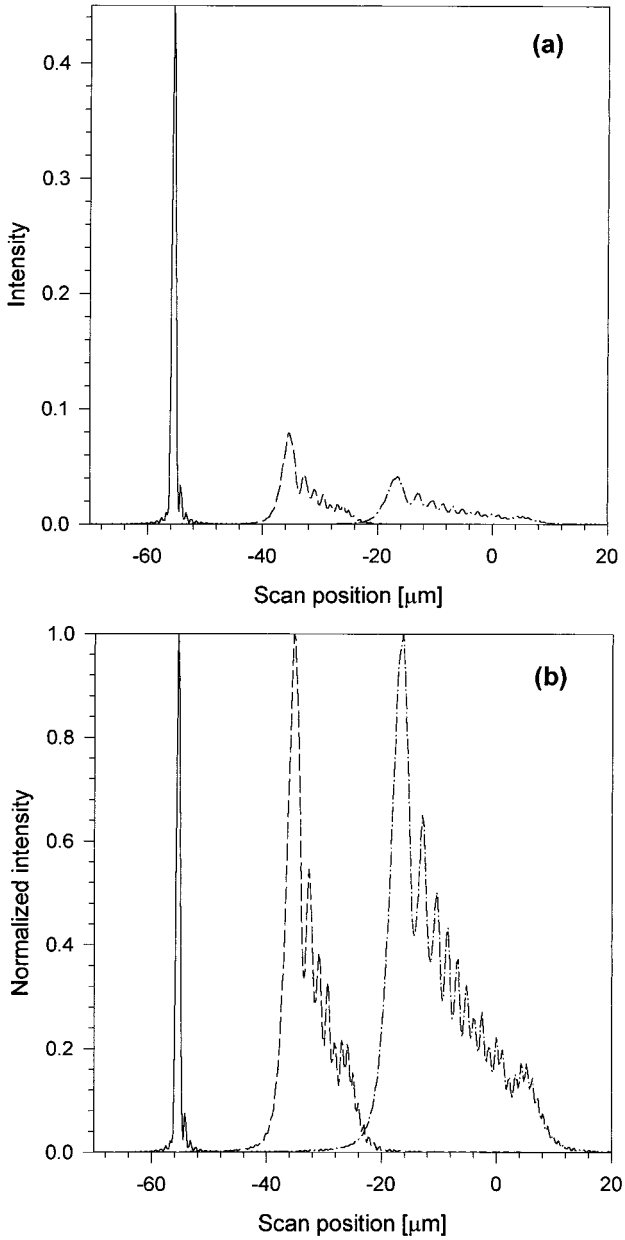


Fig. 4. Time-averaged electric-energy density distributions as functions of the scan position for a lens numerical aperture of 0.9 and for focusing from air ($n_1 = 1$) through a cover glass ($n_2 = 1.54$) into an aqueous medium ($n_3 = 1.33$). The thicknesses of the cover glasses are 120 (solid curve), 170 (dashed curve), and 220 μm (dashed-dotted curve) and the depth in the aqueous medium is 50 μm. The lens is corrected for a 170-μm cover glass.

IBM PC-compatible 90-MHz Pentium computer running the Linux operating system. Numerical results were plotted by use of the SIGMAPLOT plotting software. No post data processing was performed prior to data visualization. The origin of the coordinate for all plots presented below was set to coincide with the Gaussian focus, and the wavelength used in the computations was 488 nm.

The time-averaged electric-energy density as a function of scan position (i.e., the specimen is scanned and the horizontal axis shows the distance of the last interface from the unaberrated Gaussian focus) is shown in Fig. 3(a) for a lens that is not corrected for the cover glass. Focusing occurs from air ($n_1 = 1$) through the cover glass ($n_2 = 1.54$) and to the aqueous medium ($n_3 = 1.33$) of the specimen. Individual curves correspond to 120- (solid curve), 170- (dashed curve), and 220-μm (dashed-dotted curve) cover-glass thicknesses and a depth of 50 μm in the aqueous medium. This figure shows that the spherical aberration associated with an increasing cover-glass thickness adversely affects the distributions. With

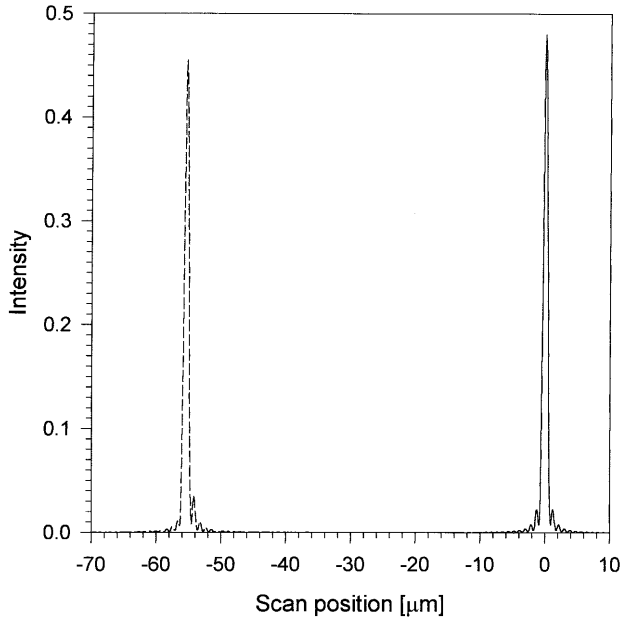


Fig. 5. Time-averaged electric-energy density distributions as functions of scan position for a lens numerical aperture of 0.9 and for focusing from air ($n_1 = 1$) through a cover glass ($n_2 = 1.54$) into an aqueous medium ($n_3 = 1.33$). The lens is corrected for a 170-μm cover glass. The solid curve corresponds to a 0-μm depth and a 170-μm cover-glass thickness, whereas the dashed curve corresponds to a 50-μm depth and a 120-μm cover-glass thickness.

increasing spherical aberration the main peak energy dramatically decreases. As shown in Fig. 3(b), in which the same distributions are plotted on a normalized scale, the full width at half-maximum (FWHM) values of the distributions also increase with increasing spherical aberration. There is a well-pronounced positive focus shift present on these distributions.

The time-averaged electric-energy density as a function of scan position is shown in Figs. 4(a) and 4(b) for a lens that is corrected for a 170-μm coverglass thickness. Focusing occurs from air ($n_1 = 1$) through the cover glass ($n_2 = 1.54$) and to the aqueous medium ($n_3 = 1.33$) of the specimen. Individual curves correspond to 120- (solid curve), 170- (dashed curve), and 220-μm (dashed-dotted curve) cover glass thicknesses and a depth of 50 μm in the aqueous medium. This figure shows that, since the lens was corrected for a 170-μm cover glass, the distributions become distorted for a depth of 50 μm in the aqueous medium. The least aberration is shown by the curve corresponding to the 120-μm cover-glass thickness. This is because the 120-μm cover glass and the additional 50 μm of the aqueous medium cause a spherical aberration slightly less than that which would be caused by a 170-μm glass and a 50-μm depth. Indeed, when we compare the distributions corresponding to a 120-μm cover glass and a 50-μm aqueous medium (dashed curve) to that of a 170-μm cover glass and 0-μm aqueous medium (solid curve), as shown in Fig. 5, it is clearly seen that, although the former exhibits an asymmetrical distribution, the main two peaks are comparable both in peak value and FWHM. We can thus conclude that imaging with an exactly corrected lens deep from an aqueous medium results in a worsened resolution compared with that obtained when a thinner cover glass is used in the experiment and the imaging occurs from greater depths of the aqueous medium. This conclusion has implications for biological microscopy and more closely for biological confocal microscopy: When the specimen (whose refractive index is usually close to that of water) is thick, the use of a thinner-than-nominal cover-glass thickness could significantly improve the resolution.

7. Conclusions

We have presented a new solution to electromagnetic diffraction for the problem of light focused into a stratified medium. We have solved the above diffraction problem for the case of a multiple-planar interface in a rigorous mathematical manner, and the solution satisfies Maxwell's equations and is therefore valid everywhere except in the immediate vicinity of the aperture. The solution can be regarded as an extension of Wolf's integral formulas for focusing into a single medium and is valid for high-aperture focusing. We have shown that, for our case, the aberration function is in a closed analytical form and the aberration introduced by the stratified medium can be simply derived from aberration caused by individual layers.

We have also presented numerical examples applicable to biological microscopy and shown that, when imaging occurs from greater depths of an aqueous medium through a coverglass, it is more advantageous to use a thinner-than-nominal coverglass to obtain better resolution.

The main advantages of our method are as follows: The solution is in a simple form that can be used directly for numerical computation. The method for obtaining the strength vectors is generally applicable. Incident electric vectors, with directions other than perpendicular to the optical axis, can readily be treated. The solution is consistent, and, for the special case of a single medium of propagation, it reduces to the results published previously. Apart from the initial assumption of Wolf's integral formulas, no approximations are used.

Appendix A

Let us denote the amplitude of a plane wave incident upon the first interface by $\mathbf{A}^{(1)}$. The amplitude of the plane wave refracted by the stratified medium is denoted by $\mathbf{B}^{(N-1)}$. The transmission coefficient of the stratified medium is defined as

$$T_m^{(N-1)} = \frac{B_m^{(N-1)}}{A_m^{(1)}}, \tag{A1}$$

where $m = s, p$ denotes the TE and TM incident waves, respectively.

The transmission coefficient of a stratified medium

is thus given by (Ref. 8, Section 1.6)

$$T_m^{(N-1)} = \frac{2p_m^{(1)}}{[K_{11}^{(N-1)} + K_{12}^{(N-1)}p_m^{(N)}]p_m^{(1)} + [K_{21}^{(N-1)} + K_{22}^{(N-1)}p_m^{(N)}]}, \tag{A2}$$

where

$$p_s^{(j)} = n_j \cos\theta_j, \qquad p_p^{(j)} = \frac{1}{n_j}\cos\theta_j, \qquad j = 1, 2, \ldots, N,$$

and $K_{11}^{(N-1)}, \ldots, K_{22}^{(N-1)}$ are the elements of matrix $\mathbf{K}^{(N-1)}$ that satisfy the following recursion relation:

$$\mathbf{K}^{(N-1)} = \mathbf{K}^{(N-2)} \cdot \mathbf{M}^{(N-1)}(h_{N-1} - h_{N-2}), \tag{A3}$$

where the elements of matrix $\mathbf{M}$ are given by

$$\mathbf{M}^{(N-1)} = \begin{bmatrix} \cos\beta_{N-1} & -\dfrac{i}{p_m^{(N-1)}}\sin\beta_{N-1} \\ -ip_m^{(N-1)}\sin\beta_{N-1} & \cos\beta_{N-1} \end{bmatrix}. \tag{A4}$$

We obtain from Eqs. (A2) and (A3)

$$\frac{2p_m^{(1)}}{T_m^{(N-1)}} = [K_{1j}^{(N-2)}M_{j1}^{(N-1)} + K_{1j}^{(N-2)}M_{j^2}^{(N-1)}p_m^{(N)}]p_m^{(1)} + [K_{2j}^{(N-2)}M_{j1}^{(N-1)} + K_{2j}^{(N-2)}M_{j^2}^{(N-1)}p_m^{(N)}], \tag{A5}$$

where at double indices a summation needs to be performed. Now, substituting Eq. (A4) into Eq. (A5) we obtain

$$T_m^{(N-1)} = \frac{t_m^{(N-1)}T_m^{(N-2)}\exp(i\beta_{N-1})}{D^{(N-2)}}, \tag{A6}$$

where

$$t_m^{(N-1)} = \frac{2p_m^{(N-1)}}{p_m^{(N-1)} + p_m^{(N)}}$$

is the transmissivity of the interface between layers $N-1$ and N and

$$D^{(N-2)} = 1 + \frac{2}{p_m^{(1)}}r_m^{(N-1)}T^{(N-2)}\{[K_{11}^{(N-2)} - K_{12}^{(N-2)}p_m^{(N-1)}]p_m^{(1)} + K_{21}^{(N-2)} - K_{22}^{(N-2)}p_m^{(N-1)}\}\exp(2i\beta_{N-1}), \tag{A7}$$

where

$$r_m^{(N-1)} = \frac{p_m^{(N-1)} - p_m^{(N)}}{p_m^{(N-1)} + p_m^{(N)}},$$

from which Eq. (A6) can be written as

$$T_m^{(N-1)} = \frac{t_m^{(N-1)}\prod_{j=1}^{N-2} t_m^{(j)}\exp(i\beta_{j+1})}{D_m^{(N-1)}}, \tag{A8}$$

where

$$D_m^{(2)} = 1 + r_m^{(1)}r_m^{(2)}\exp(2i\beta_2),$$

$$D_m^{(3)} = 1 + r_m^{(1)}\{r_m^{(2)}\exp(2i\beta_2) + r_m^{(3)}\exp[2i(\beta_2 + \beta_3)]\} + r_m^{(2)}r_m^{(3)}\exp(2i\beta_3),$$

$$D_m^{(4)} = 1 + r_m^{(1)}\{r_m^{(2)}\exp(2i\beta_2) + r_m^{(3)}\exp[2i(\beta_2 + \beta_3)] + r_m^{(4)}\exp[2i(\beta_2 + \beta_3 + \beta_4)]\} + r_m^{(2)}\{r_m^{(3)}\exp(2i\beta_3) + r_m^{(4)}\exp[2i(\beta_3 + \beta_4)]\} + r_m^{(3)}r_m^{(4)}\exp(2i\beta_4) + r_m^{(1)}r_m^{(2)}r_m^{(3)}r_m^{(4)}\exp[2i(\beta_2 + \beta_4)]. \tag{A9}$$

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